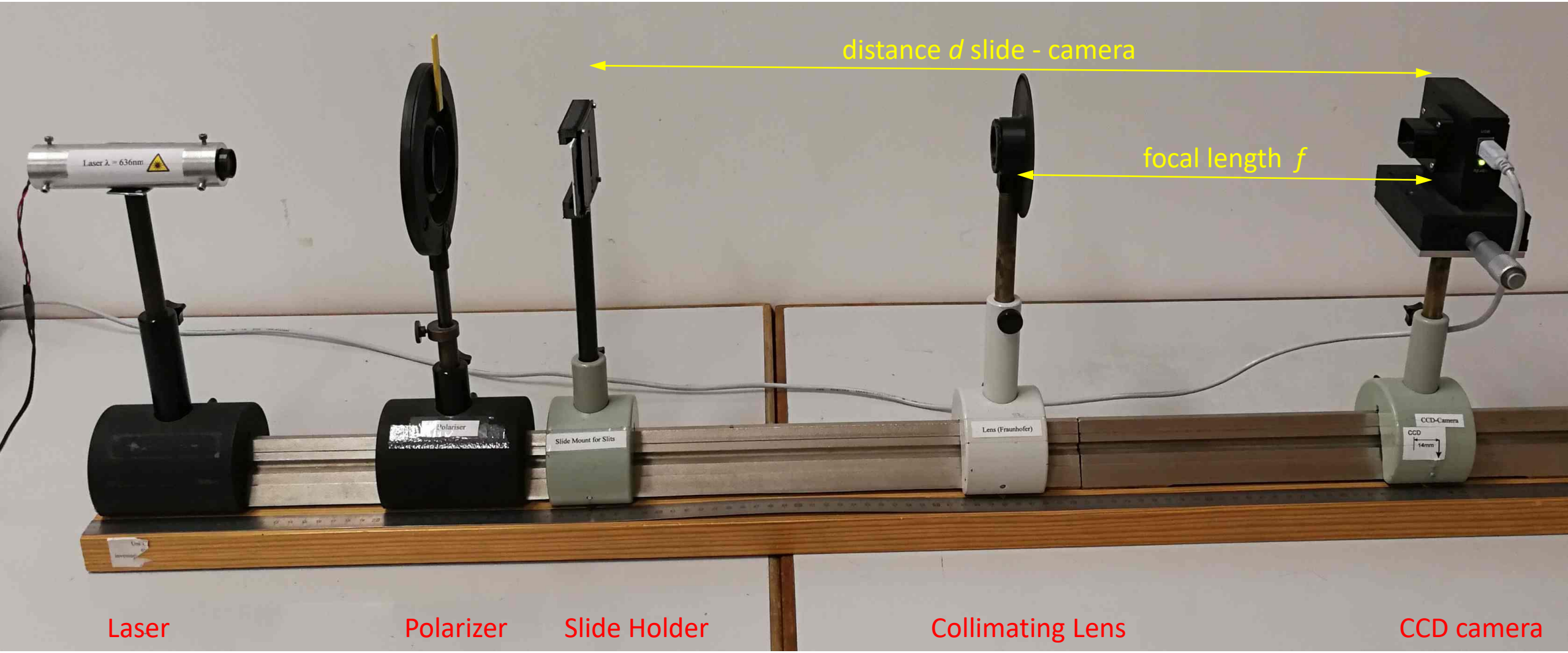
Guide

to

Experiment O17e: Diffraction

Setup: Fraunhofer diffraction

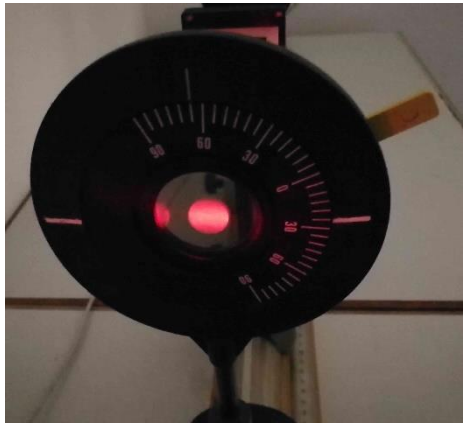


The laser light is linearly polarized.

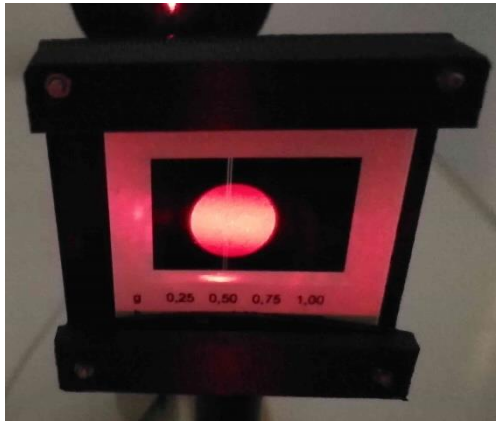
Sometimes two polarizers are necessary to properly reduce the light intensity.

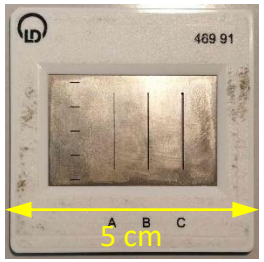
In the slide holder thin plates with single, double, triple and quadruple slits are inserted.

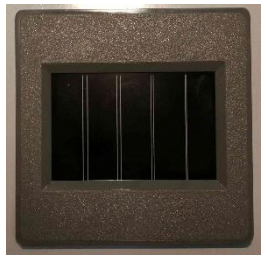
Polarizer

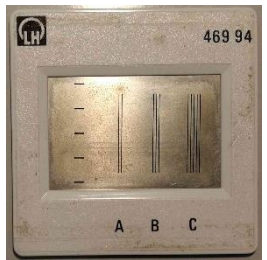


Slide holder
with slides



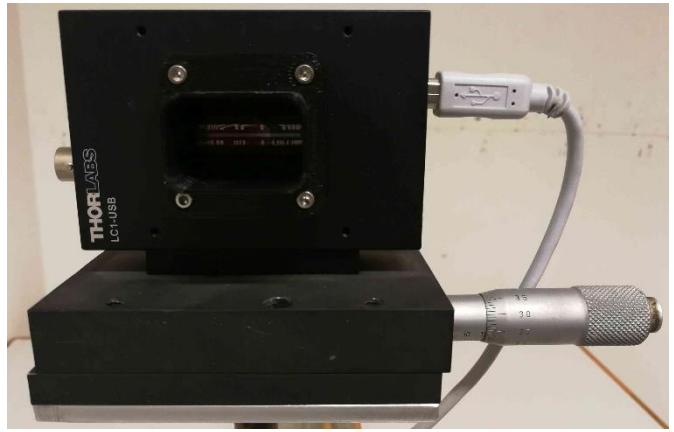






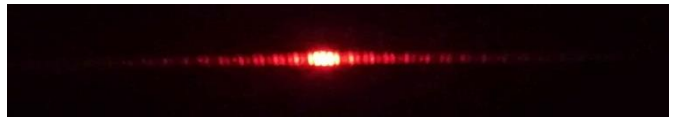
Lens

Camera



Camera with micrometer screw for controlled horizontal movement perpendicular to the optical axis.

Illuminated camera.



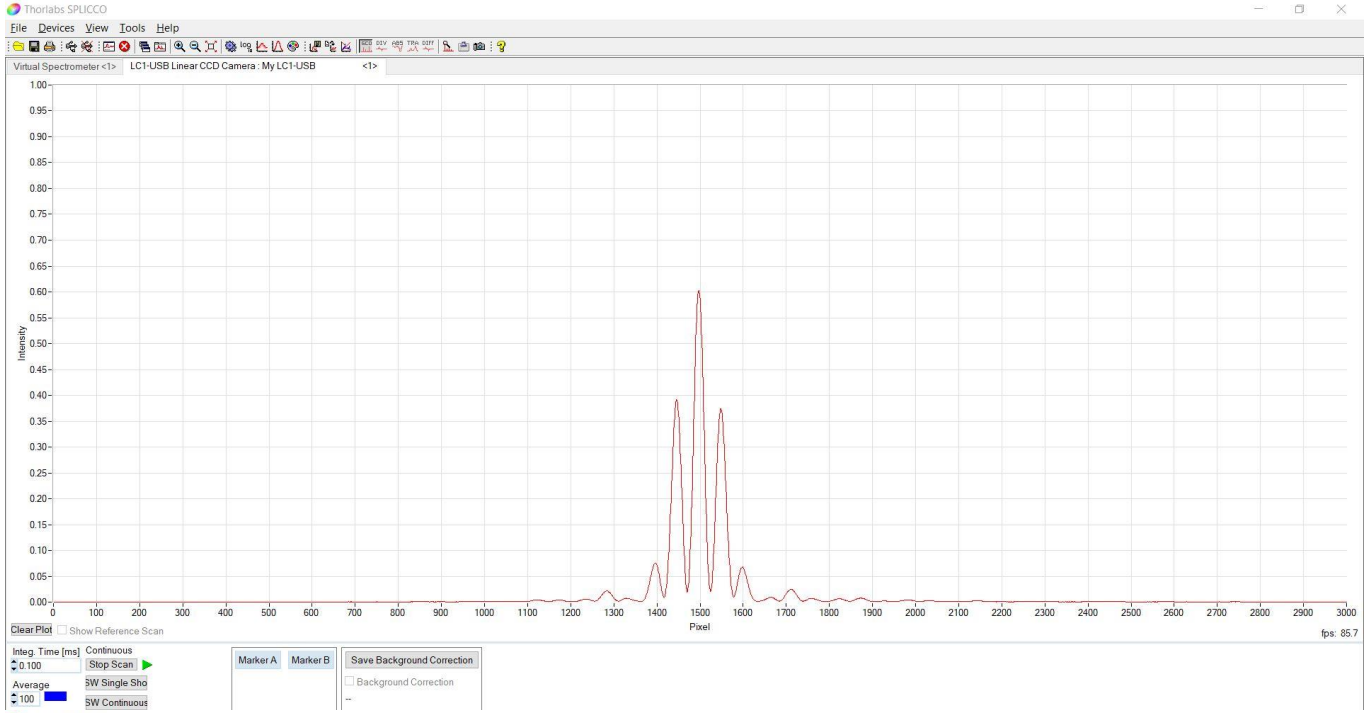
Double slit intensity pattern on paper.

As always in an optics experiment the various optical components on the optical bench have to be carefully adjusted in alignment and in height.
The height of the camera is especially critical.

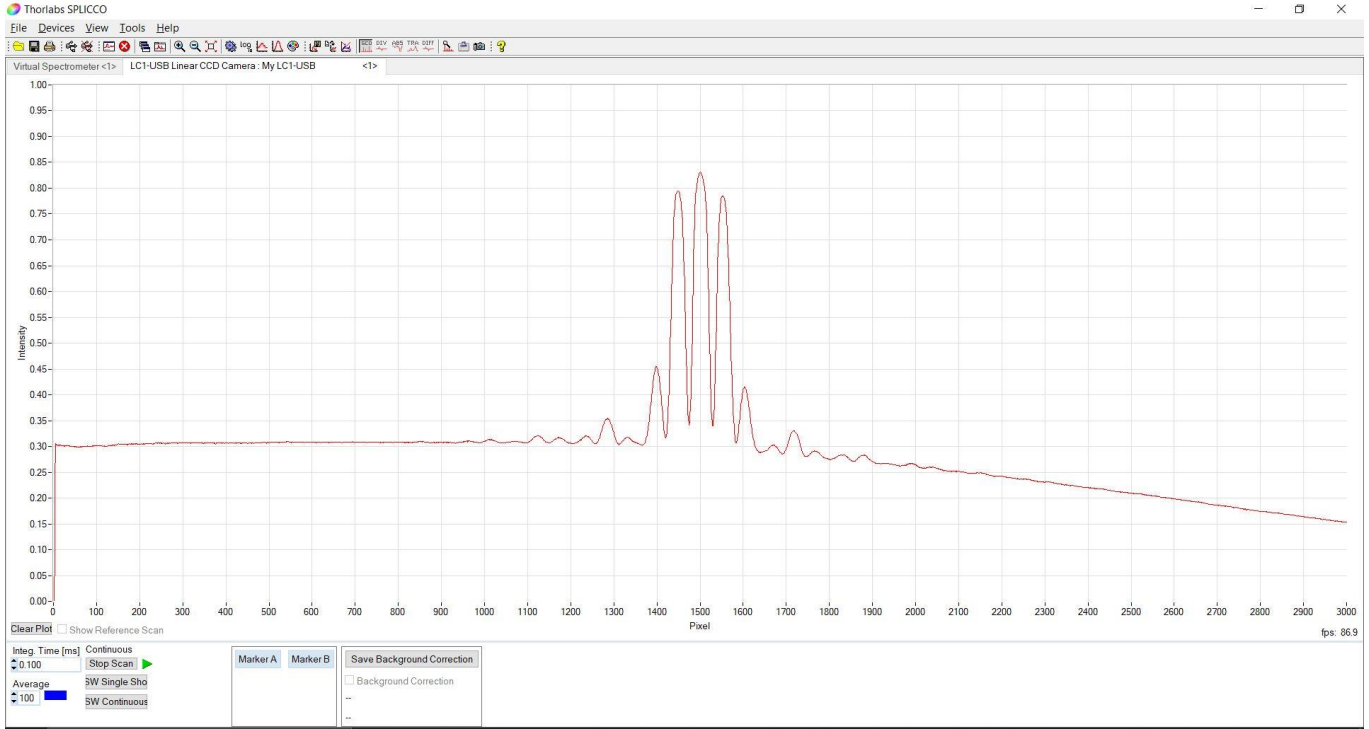
Software

The camera is controlled by a software called “Splicco”, see screenshots to the right.
Mainly, the software allows to set integration time (here: 0.1 ms) and number of averages (here: 100).
The data are saved into a csv-file with two columns: x-column is pixel number from 1 to 3000, y-column is intensity in arbitrary units.

On the screenshot the diffraction pattern looks quite featureless. Have a look on it on a semi-logarithmic scale.



Double slit intensity profile recorded in dark lab.

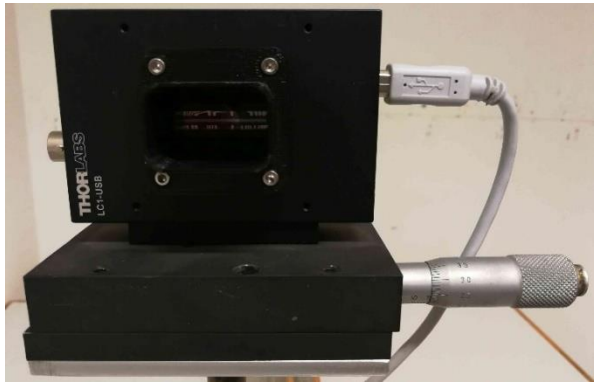


The same profile with the ceiling lamps on.

Camera Specifications

The cameras are CCD line cameras with a single line with either 3000 or 2048 pixels. The pixels are rectangular with the sizes specified below. The smaller dimension is the horizontal size which you need to convert the image into physical dimensions.

Old Camera



New Camera

Optical Head:

Spectral Range:	350 to 1000 nm
CCD Integration Time:	1 µs to 200 ms
CCD Sensitivity:	300 V/(lx·s)
CCD Pixel Size:	7 µm x 200 µm (7 µm pitch)
Dimensions (LxWxH):	3.6x2.6x1.0 Inches
Weight:	1 lb

Sensor Specifications	
Detector Range (CCD Chip)	350 - 1100 nm
CCD Pixel Size	14 µm x 56 µm (14 µm pitch)
CCD Sensitivity	240 V / (lx · s)
CCD Dynamic Range [1])	333
CCD Pixel Number	2048
Integration Time	1.054 ms – 50 s